Lab 6: Code and Branch Coverage Analysis

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Sec: F
Modules: salon_booking.py (module under test), CodeCoverageTest_salon_booking.py (code coverage tests), BranchCoverageTest_salon_booking.py (branch coverage tests)

Code Coverage

```
(venv) piyush-a-patel@piyush:/mnt/windows/Piyush/Programs/SE/lab6/L6_L7_42_salon_booking/Lab 6$ pip install coverage pytest
coverage run -m pytest CodeCoverageTest_salon_booking.py
coverage report -m
Requirement already satisfied: coverage in /mnt/windows/Piyush/Programs/SE/lab6/venv/lib/python3.12/site-packages (7.17.0)
Requirement already satisfied: pytest in /mnt/windows/Piyush/Programs/SE/lab6/venv/lib/python3.12/site-packages (9.2.0)
Requirement already satisfied: iniconfig>=2 in /mnt/windows/Piyush/Programs/SE/lab6/venv/lib/python3.12/site-packages (from pytest) (2.3.1)
Requirement already satisfied: packaging>=24 in /mnt/windows/Piyush/Programs/SE/lab6/venv/lib/python3.12/site-packages (from pytest) (26.3)
Requirement already satisfied: pluggy<2,>=1.6 in /mnt/windows/Piyush/Programs/SE/lab6/venv/lib/python3.12/site-packages (from pytest) (1.7.0)
Requirement already satisfied: pygments>=2.15 in /mnt/windows/Piyush/Programs/SE/lab6/venv/lib/python3.12/site-packages (from pytest) (2.21.0)
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-9.2.0, pluggy-1.7.0
rootdir: /mnt/windows/Piyush/Programs/SE/lab6/L6_L7_42_salon_booking/Lab 6
collected 14 items

CodeCoverageTest_salon_booking.py ..............                         [100%]

============================== 14 passed in 0.10s ==============================
Name                               Stmts   Miss  Cover   Missing
----------------------------------------------------------------
CodeCoverageTest_salon_booking.py      32      0   100%
salon_booking.py                       37      0   100%
----------------------------------------------------------------
(venv) piyush-a-patel@piyush:/mnt/windows/Piyush/Programs/SE/lab6/L6_L7_42_salon_booking/Lab 6$
```

Branch Coverage

```
(venv) piyush-a-patel@piyush:/mnt/windows/Piyush/Programs/SE/lab6/L6_L7_42_salon_booking/Lab 6$ coverage erase
coverage run --branch -m pytest BranchCoverageTest_salon_booking.py
coverage report -m
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-9.2.0, pluggy-1.7.0
rootdir: /mnt/windows/Piyush/Programs/SE/lab6/L6_L7_42_salon_booking/Lab 6
collected 27 items

BranchCoverageTest_salon_booking.py ...........................                 [100%]

============================== 27 passed in 0.16s ==============================
Name                                Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------------------------
BranchCoverageTest_salon_booking.py      71      0      0      0   100%
salon_booking.py                         37      0     26      0   100%
-----------------------------------------------------------------------------
TOTAL                                   108      0     26      0   100%
(venv) piyush-a-patel@piyush:/mnt/windows/Piyush/Programs/SE/lab6/L6_L7_42_salon_booking/Lab 6$
```

Observations

1. Code coverage: 14 tests passed and salon_booking.py reached 100% line coverage (37 of 37 statements), above the 90% requirement.
2. Branch coverage: 27 tests passed and salon_booking.py reached 100% branch coverage (26 of 26 branches, 0 partial), above the 90% requirement.
3. The starter test only covered a short haircut and one valid booking, so most branches were never run. Line coverage alone would miss the False side of each condition; the branch tests close that gap.
4. Boundary values were tested explicitly: exactly the included time, 60 and 61 minutes of overrun, walk-in party sizes 2 and 3, and the open and late cutoffs for colour and weekend bookings.